Alexandre **Pasco**

Postdoctoral Researcher · Ph.D. Applied Mathematics · Graduated Engineer

EPFL SB MATH HPNALGS, CH-1015 Lausanne, Switzerland

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My research focuses on dimension reduction, randomized methods for high-dimensional approximation and inverse problems, and algebraic domain decomposition methods.
I am also committed to promoting open science.

Education

Ph.D. applied mathematics, Ecole Centrale de Nantes — *Nantes, France* — *Dec. 2022 - Jan. 2026*

(Master's) Engineer's degree: applied mathematics, Ecole Centrale de Nantes — *Nantes, France* — *Sep. 2019 - Sep. 2022*

Prepa Mathematics-Physics, Lycée Clémenceau — *Nantes, France* — *Sep. 2017 - Aug. 2019*

Experience

Postdoctoral Researcher, École Polytechnique Fédérale de Lausanne, Chair HPNALGS — *Lausanne, Switzerland* — *Apr. 2026 - Present*

Teacher assistant, École Centrale de Nantes — *Nantes, France* — *Feb. 2023 - Mar. 2025*

Researcher internship, Laboratoire Jean Leray — *Nantes, France* — *Apr. 2022 - Sep. 2022*

Actuarial internship, Union Mutualiste Retraite — *Nantes, France* — *Apr. 2021 - Aug. 2021*

Publications

Journal articles

- Nouy, A. and Pasco, A.: Surrogate to Poincaré inequalities on manifolds for dimension reduction in nonlinear feature spaces. — *Mar. 2026*
  Information and Inference: A Journal of the IMA. https://doi.org/10.1093/imaiai/iaag005
- Nouy, A. and Pasco, A.: Dictionary-based model reduction for state estimation. — *Apr. 2024*
  Advances in Computational Mathematics. https://doi.org/10.1007/s10444-024-10129-4

Prepublications

- Pasco, A. and Nouy, A.: Gradient-based collective and structured nonlinear dimension reduction with quadratic surrogates. — *Feb. 2026*
  Submitted. https://arxiv.org/abs/2602.01143

Open software

- Nouy, A. and Pasco, A.: Poincaré inequality based nonlinear dimension reduction — *Nov. 2025*
  Tensap v1.8. https://github.com/anthony-nouy/tensap/releases/tag/v1.8

Other

- Pasco, A.: Nonlinear dimension reduction for high dimensional approximation and inverse problems. — *Jan. 2026*
  HAL. https://hal.science/tel-05469205v1

Presentations

Oral presentations

- Séminaire des doctorants du LMJL — *Nantes, France*
  Model Reduction for parameterized PDEs: why and how ? — *Mar. 2023*
- Young Mathematicians in Model Order Reduction Conference (YMMOR) — *Ulm, Germany*
  Dictionary-based model reduction for state estimation. — *Mar. 2023*
- Séminaire de Mathématiques Appliquées du LMJL — *Nantes, France*
  Dictionary-based model reduction for state estimation. — *Apr. 2023*
- Congrès des Jeunes Chercheuses et Chercheurs en Mathématiques et Applications (CJC-MA) — *Gif-sur-Yvette, France*
  Dictionary-based model reduction for state estimation. — *Sep. 2023*

- SIAM Conference on Uncertainty Quantification — *Trieste, Italy*
  Dictionary-based model reduction for state estimation. — *Feb. 2024*
- Young Mathematicians in Model Order Reduction Conference (YMMOR) — *Stuttgart, Germany*
  Structured Nonlinear Dimension Reduction Using Gradient Evaluations. — *Mar. 2024*
- Rencontres Doctorales Lebesgue (RDL) — *Angers, France*
  Dimension reduction using Poincaré inequalities. — *Apr. 2024*
- Semaines d'Etudes Mathematiques - Entreprises et Societe (SEMES). — *Nantes, France*
  Reconstruction of a latent space from a redox signal. — *Nov. 2024*
- Séminaire des jeunes chercheurs en analyse de l'IRMAR. — *Rennes, France*
  Poincaré inequalities on manifolds for dimension reduction in nonlinear feature spaces. — *Dec. 2024*
- Young Mathematicians in Model Order Reduction Conference (YMMOR) — *Trieste, Italy*
  Surrogate to Poincaré inequalities on manifolds for dimension reduction in nonlinear feature spaces. — *May. 2025*
- RT-UQ PhD-day — *Grenoble, France*
  Surrogate to Poincaré inequalities on manifolds for dimension reduction in nonlinear feature spaces. — *May. 2025*
- PhD Defense — *Nantes, France*
  Nonlinear dimension reduction for high dimensional approximation and inverse problems. — *Jan. 2026*
- Congrès des Jeunes Chercheuses et Chercheurs en Mathématiques et Applications (CJC-MA) — *Paris, France*
  Random sketching of operators and application to learning preconditioners for model order reduction. — *Mar. 2026*
- 11th International Conference on Curves and Surfaces — *Saint-Malo, France*
  Surrogate to Poincaré inequalities on manifolds for collective dimension reduction in nonlinear feature spaces. — *Jun. 2026*

Poster presentations

- Mascot-Num Conference — *Le Croisic, France*
  Dictionary-based model reduction for state estimation. — *Apr. 2023*
- 46ème Congrès National d'Analyse Numérique (CANUM) — *Ile de Ré, France*
  Dictionary-based model reduction for state estimation. — *May. 2024*
- RandNLA workshop — *Lausanne, Switzerland*
  Random sketching of matrices with application to learning preconditioners. — *Jun. 2026*

Services

Committee member, YMMOR committee — *Sep. 2023 - Present*
Doctoral student representative, Laboratoire Jean Leray laboratory board — *Nantes, France* — *Sep. 2024 - Jun 2025*
Referent Researcher in middle schools, MATh.en.JEANS — *Nantes, France* — *Oct. 2023 - Mar 2025*
Volunteer tutor in Maths and Physics, Ecole Centrale de Nantes — *Nantes, France* — *Oct. 2019 - Apr 2021*

Teaching

Teacher assistant

Mathematics for engineering, Ecole Centrale de Nantes, Undergraduate course — *Nantes, France* — *Feb. 2023 - Jan 2025*
Ordinary Differential Equations, Ecole Centrale de Nantes, Undergraduate course — *Nantes, France* — *Feb. 2025 - Mar 2025*
Applied mathematics project, Ecole Centrale de Nantes, Graduate project supervision — *Nantes, France* — *Oct. 2023 - Apr 2025*

Skills

Programming **Python** (Advanced), **LaTeX** (Advanced), **MATLAB** (Intermediate), C++ (Basic), F90 (Basic), SQL (Basic), SAS (Basic)
Languages **French** (Native), **English** (Advanced), **Japanese** (Elementary JLPT N4), **German** (Beginner).